

Unit - III

- Sequence & Series.
- Limit of Sequence.
- test of Sequence [Convergence, Divergence, oscillations].
- Infinite Series.

test of convergence → power Series.

→ comparison test.

→ Ratio test.

→ Raabe's test.

→ Cauchy Root test.

→ logarithm test.

→ Cauchy Integral test.

- Taylor and Maclaurin Series.
- Conversion of Taylor Series.

Sequence.

A function $f: \mathbb{N} \rightarrow \mathbb{R}$ whose domain is the set of the Natural number and range is the set of the Real Number is called a sequence of a real number or simply a real sequence.

It is denoted by $\langle a_n \rangle$, $\{x_n\}$.

$$a_n = (-1)^n \quad a_1 = -1, a_2 = 1, a_3 = -1, \dots$$

$$\langle a_n \rangle = \langle -1, 1, -1, 1, \dots \rangle$$

$$a_n = \frac{1}{n}$$

$$a_1 = 1, a_2 = \frac{1}{2}, a_3 = \frac{1}{3}, \dots$$

$$\langle a_n \rangle = \langle 1, \frac{1}{2}, \frac{1}{3}, \dots \rangle$$

- A Sequence $\{x_n\}$ is said to be bounded above if there exist a real number $K: x_n \leq K$ for all natural number $n \rightarrow \infty$. The number K is called an upper Bound of the Sequence.

$$x_n = \frac{1}{n}$$

$$x_n = \left\langle 1, \frac{1}{2}, \frac{1}{3}, \dots \right\rangle$$

$$x_n \geq 1$$

↳ above Bound.

A Sequence $\{x_n\}$ is said to be bounded below if there exist a real number $K: K \leq x_n$.

$$x_n = \frac{1}{n}$$

$$x_n = \left\langle 1, \frac{1}{2}, \frac{1}{3}, \dots, 0 \right\rangle$$

$$0 \leq x_n$$

↳ lower Bound

A Sequence which is bounded both above & below is called a bounded Sequence.

$$0 < x_n = \frac{1}{n} < 1$$

$$\langle x_n \rangle = \langle 2^n \rangle$$

$$\langle x_n \rangle = \langle 2, 4, 8, 16, \dots \rangle$$

$$2 \leq \langle x_n \rangle$$

↳ Bound (lower).

Convergence of a Sequence.

A sequence $\{x_n\}$ is said to be convergent to the limit 'L' if for every $\epsilon > 0$ there exist a +ve int. n_0 : $|x_n - L| < \epsilon$ for all $n > n_0$.

If the limit of a sequence is finite the sequence is convergent.
If the limit of a sequence does not tend to a finite number the sequence is said to be divergent.

$$\langle x_n \rangle = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0 \rightarrow \text{convergent sequence.}$$

$$\langle x_n \rangle = 2n + 1$$

$$x_n = \langle 3, 5, 7, 9, \dots \rangle$$

$$\lim_{n \rightarrow \infty} x_n = \infty \rightarrow \text{divergent sequence.}$$

$$\langle x_n \rangle = \frac{n+1}{n}$$

$$\lim_{n \rightarrow \infty} x_n = 1 + \frac{1}{n} = 1 + \frac{1}{\infty} = 1 \rightarrow \text{convergent sequence.}$$

$$\langle x_n \rangle = \frac{n^2}{n!}$$

$$\lim_{n \rightarrow \infty} x_n = \frac{n^2}{n \times (n-1)!} = \frac{n}{(n-1)(n-2)(n-3)\dots 1!} = \frac{1}{(1-\frac{1}{n})(1-\frac{2}{n})\dots} = 1 \rightarrow \text{convergent sequence}$$

$$\langle x_n \rangle = \frac{n}{n+1}$$

$$\lim_{n \rightarrow \infty} x_n = \frac{n}{n+1} \rightarrow \text{convergent sequence}$$

$$\langle x_n \rangle = (-1)^n$$

$$\lim_{n \rightarrow \infty} x_n = (-1)^\infty \rightarrow \text{oscillating}$$

Monotonic Sequence:

A sequence is either increasing or decreasing sequence is called the Monotonic Sequence.

A sequence is monotonic & bounded is a convergent sequence.

$1, 4, 9, 16, \dots \rightarrow$ Monotonic.
 $-1, 1, -1, 1, \dots \rightarrow$ Non-Monotonic.
 $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \rightarrow$ Monotonic + Bounded
 $\hookrightarrow$ Convergent.

Remember Me:-

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\lim_{n \rightarrow \infty} x^n = 0 \text{ if } x < 1.$$

$$\lim_{n \rightarrow \infty} \frac{\log n}{n} = 0.$$

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0, \forall x$$

$$\lim_{n \rightarrow \infty} x^n = \infty \text{ if } x > 1.$$

$$\lim_{n \rightarrow \infty} \left[1 + \frac{1}{n}\right]^n = e$$

$$\lim_{n \rightarrow \infty} (n!)^{1/n} = \infty$$

$$\lim_{n \rightarrow \infty} [n!]^{1/n} = \infty$$

$$\lim_{n \rightarrow \infty} \left[\frac{n!}{n^n}\right]^{1/n} = \frac{1}{e}.$$

Show the Sequence whose n^{th} term is $x_n = \frac{2n-1}{3n+2}$.

$$-1, -\frac{3}{8}, -\frac{1}{11} \rightarrow \text{increasing}$$

$$\lim_{n \rightarrow \infty} \frac{2n-1}{3n+2} = \frac{2-1/n}{3+2/n} = \frac{2}{3} \rightarrow \text{convergent sequence.}$$

$$\lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{n^2} = \frac{n(n+1)}{2 \cdot n^2} = \frac{(1+\frac{1}{n})}{2} = \frac{1}{2} \rightarrow \text{convergent sequence.}$$

Series:- An expression of the form $a_1 + a_2 + a_3 + \dots + a_n$ denoted by $\sum_{n=1}^{\infty} a_n$ is called the infinite series, the term a_n is called the n^{th} term of the series where $S_n = a_1 + a_2 + \dots + a_n$ is called the partial sum of the series.

• Behaviour of the Series:-

→ Convergent:- When $\lim_{n \rightarrow \infty} S_n = \text{finite}$ then it's called the convergent series.

→ Divergent:- When $\lim_{n \rightarrow \infty} S_n = \text{infinite}$ then it's called the divergent series.

→ Oscillatory:- When $\lim_{n \rightarrow \infty} S_n = \text{more than 1 limit / is not unique limit}$.

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots \quad a_n = \frac{1}{n(n+1)} \rightarrow \text{convergent}$$

$$1^2 + 2^2 + 3^2 + 4^2 + \dots \quad a_n = n^2 \rightarrow \text{divergent}$$

$$-1 + 1 - 1 + 1 - \dots \rightarrow \text{oscillatory}$$

Geometry Properties:-

$$1 + r + r^2 + r^3 + \dots \rightarrow \infty \text{ where } |r| < 1 \rightarrow \text{convergent}$$

$$|r| > 1 \rightarrow \text{divergent}$$

$$1 + \frac{1}{3} + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \dots \rightarrow \infty \rightarrow \text{convergent}$$

$$\# \frac{2}{1 \cdot 3} + \frac{5}{3 \cdot 5} + \frac{10}{5 \cdot 7} + \dots$$

$$a_n = \frac{n^2+1}{(2n-1)(2n+1)}$$

$$S = \sum_{n=1}^{\infty} \frac{n^2+1}{(2n-1)(2n+1)}$$

$$\lim_{n \rightarrow \infty} \frac{n^2+1}{(2n-1)(2n+1)}$$

$$\lim_{n \rightarrow \infty} \frac{n^2+1}{4n^2-1}$$

$$\lim_{n \rightarrow \infty} \frac{n^2(1 + \frac{1}{n^2})}{n^2(4 - \frac{1}{n^2})} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{4} = \frac{1}{4}$$

$\frac{1}{4} < 1 \rightarrow$ divergent.

• Test of the Series:-

→ Ratio test (D'Alembert's test):-

Let $\sum_{n=1}^{\infty} a_n$ or $\sum_{n=1}^{\infty} a_n$ be a series of positive terms.

i) if $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} > 1 \rightarrow$ divergent.

ii) if $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} < 1 \rightarrow$ convergent.

iii) if $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1 \rightarrow$ Ratio test fail.

$$\# 1 + \frac{1^2 \cdot 2^2}{1 \cdot 3 \cdot 5} + \frac{1^2 \cdot 2^2 \cdot 3^2}{1 \cdot 3 \cdot 5 \cdot 7 \cdot 9} + \dots$$

$$a_n = \frac{(n!)^2}{1 \cdot 3 \cdot 5 \dots (2n+1)}$$

$$a_{n+1} = \frac{[(n+1)!]^2}{1 \cdot 3 \cdot 5 \dots (2n+1)(2n+3)}$$

Applying the Ratio test.

$$\lim_{n \rightarrow \infty} \left(\frac{a_{n+1}}{a_n} \right)$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^2}{2n+3}$$

$$\lim_{n \rightarrow \infty} \frac{n^2 + 2n + 1}{2n + 3}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n^2}{n} + \frac{2n}{n} + \frac{1}{n}}{\frac{2n}{n} + \frac{3}{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{n + 2 + \frac{1}{n}}{2 + \frac{3}{n}}$$

$$\frac{\infty + 2 + 0}{2 + 0} = \infty$$

$\infty > 1 \rightarrow$ divergent

$\frac{x^n}{n!}$

$$a_n = \frac{x^n}{n!}$$

$$a_{n+1} = \frac{x^{n+1}}{(n+1)!}$$

$$\frac{a_{n+1}}{a_n} \Rightarrow \frac{x^{n+1}}{(n+1)!} \times \frac{n!}{x^n}$$

$$\Rightarrow \frac{x}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{x}{n+1} = 0$$

$0 < 1 \rightarrow$ convergent.

$\left[\frac{1}{3}\right]^2 + \left[\frac{1 \cdot 2}{3 \cdot 5}\right]^2 + \left[\frac{1 \cdot 2 \cdot 3}{3 \cdot 5 \cdot 7}\right]^2 - \dots$

$$a_n = \left[\frac{1 \cdot 2 \cdot 3 \dots n}{3 \cdot 5 \cdot 7 \dots (2n+1)} \right]^2 = \left[\frac{n!}{3 \cdot 5 \cdot 7 \dots (2n+1)} \right]^2$$

$$a_{n+1} = \left[\frac{(n+1)!}{3 \cdot 5 \cdot 7 \dots (2n+1)(2n+3)} \right]^2$$

$$\frac{a_{n+1}}{a_n} = \left[\frac{(n+1)!}{3 \cdot 5 \cdot 7 \dots (2n+1)(2n+3)} \right]^2 \times \left[\frac{3 \cdot 5 \cdot 7 \dots (2n+1)}{n!} \right]^2$$

$$\frac{a_{n+1}}{a_n} = \left[\frac{n+1}{2n+3} \right]^2$$

$$\lim_{n \rightarrow \infty} \left[\frac{n+1}{2n+3} \right]^2$$

$$\lim_{n \rightarrow \infty} \left[\frac{1 + \frac{1}{n}}{2 + \frac{3}{n}} \right]^2 = \left(\frac{1+0}{2+0} \right)^2 = \left(\frac{1}{2} \right)^2$$

$$\lim_{n \rightarrow \infty} = \frac{1}{4}$$

$\frac{1}{4} < 1 \rightarrow$ convergent.

$$\# \quad 1 + \frac{x}{2} + \frac{x^2}{5} + \frac{x^3}{10} + \dots \quad \#$$

$$a_n = \frac{x^{n-1}}{(n-1)^2 + 1}$$

$$a_{n+1} = \frac{x^n}{n^2 + 1}$$

$$\frac{a_{n+1}}{a_n} = \frac{x^n}{n^2 + 1} \times \frac{(n-1)^2 + 1}{x^{n-1}}$$

$$= \frac{x \cdot ((n-1)^2 + 1)}{n^2 + 1}$$

By applying limit $\lim_{n \rightarrow \infty}$

$$\lim_{n \rightarrow \infty} x \frac{(n-1)^2 + 1}{n^2 + 1}$$

$$x \lim_{n \rightarrow \infty} \frac{(n-1)^2 + 1}{n^2 + 1}$$

$$x \cdot \lim_{n \rightarrow \infty} \frac{n^2 - 2n + 1 + 1}{n^2 + 1}$$

$$x \cdot \lim_{n \rightarrow \infty} \frac{n^2 - 2n + 2}{n^2 + 1} \Rightarrow x \lim_{n \rightarrow \infty} \frac{1 - \frac{2}{n} + \frac{2}{n^2}}{1 + \frac{1}{n^2}}$$

$$x \lim_{n \rightarrow \infty} 1 = x$$

if $x < 1 \rightarrow$ convergent.

$x > 1 \rightarrow$ divergent.

Comparison test:- If $\sum u_n$ & $\sum v_n$ are two series of positive terms if $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \text{finite and non-zero}$ then $\sum u_n$ & $\sum v_n$ both are converges or diverges together.

P-Series test:- $1 + \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \dots + \frac{1}{n^p}$

if $p > 1 \rightarrow \text{convergent}$
 $p \leq 1 \rightarrow \text{divergent}$

$\sum \frac{(n+1)(n+2)}{(n^2+1)(n^2+2)}$

$$\sum \frac{n^2+3n+2}{n^4+3n^2+2}$$

$$\sum \frac{n^2(1+3/n+2/n^2)}{n^4(1+3/n^2+2/n^4)}$$

$$v_n = 1/n^2$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \frac{(n+1)(n+2)}{(n^2+1)(n^2+2)} \cdot \frac{1}{1/n^2}$$

$$\Rightarrow \frac{n^2(n^2+3n+2)}{n^4+3n^2+2}$$

$$\Rightarrow \frac{n^4+3n^3+2n^2}{n^4+3n^2+2}$$

$$\lim_{n \rightarrow \infty} \frac{1 + 3/n + 2/n^2}{1 + 3/n^2 + 2/n^4} \Rightarrow \frac{1+0+0}{1+0+0} = 1$$

By comparison test, both the series will behave same.

and by p-series test $1/n^2 = 2 > 1$

the both series will converge.

$\sum \frac{2n^3+1}{n^5+1}$

$$\sum \frac{n^3(2+\frac{1}{n^3})}{n^5(1+\frac{1}{n^5})}$$

$$u_n = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \frac{n^3(2+\frac{1}{n^3})}{n^5(1+\frac{1}{n^5})} \times n^2$$

$$\Rightarrow \frac{(2+\frac{1}{n^3})}{(1+\frac{1}{n^5})} \text{ on applying limit.}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = 2 \text{ by comparison test, both will behave same}$$

$$\text{and } p\text{-Series test } \frac{1}{n^2} \Rightarrow 2 > 1$$

the both series will converge.

$\sum \frac{x^n}{(2n)!}$

$$u_n = \frac{x^n}{(2n)!}$$

$$u_{n+1} = \frac{x^{n+1}}{(2n+2)!}$$

$$\frac{u_{n+1}}{u_n} = \frac{x^{n+1}}{(2n+2)!} \times \frac{(2n)!}{x^n}$$

$$\frac{u_{n+1}}{u_n} = \frac{(2n)! \times x}{(2n+2)(2n+1)(2n)!}$$

$$\frac{u_{n+1}}{u_n} = \frac{x}{(2n+2)(2n+1)}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 0 < 1$$

convergent.

Cauchy Root test: if $\sum a_n$ be a series of positive term, then
 i) the series $\sum a_n$ is convergent if

$$\lim_{n \rightarrow \infty} (a_n)^{1/n} < 1.$$

ii) the series $\sum a_n$ is divergent if

$$\lim_{n \rightarrow \infty} (a_n)^{1/n} > 1.$$

iii) When $\lim_{n \rightarrow \infty} (a_n)^{1/n} = 1$ then root test fails.

$\frac{2}{3}x + \frac{8^2}{2^3}x^2 + \frac{4^3}{3^4}x^3 \dots$

$$a_n = \frac{(n+1)^n x^n}{n^{(n+1)}}$$

$$(a_n)^{1/n} = \frac{n+1}{n(n)^{1/n}} x$$

$$\rightarrow \frac{n(1+\frac{1}{n})}{n(n)^{1/n}} x$$

$$\frac{1}{1} x = x.$$

$x < 1 \rightarrow$ convergent.

$x > 1 \rightarrow$ divergent.

$x = 1 \rightarrow$ test fails.

$\frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4}\right)^2 x^2 + \left(\frac{4}{5}\right)^3 x^3 \dots$

$$a_n = \left[\frac{n+1}{n+2}\right]^n x^n.$$

$$(a_n)^{1/n} = \left[\left[\frac{n+1}{n+2}\right]^n x^n\right]^{1/n}$$

$$= \frac{n+1}{n+2} x.$$

$$\lim_{n \rightarrow \infty} \frac{n+1}{n+2} \times x.$$

$$x \cdot \lim_{n \rightarrow \infty} \frac{n+1}{n+2}$$

$$x \cdot \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{1 + \frac{2}{n}}$$

$$x \cdot \frac{1+0}{1+0} = x \times 1 = x.$$

$x < 1 \rightarrow$ convergent.

$x > 1 \rightarrow$ divergent.

$x = 1 \rightarrow$ test fails.

$$\# \quad 1 + \frac{x}{2} + \left(\frac{x}{3}\right)^3 + \left(\frac{x}{4}\right)^3 \dots$$

$$a_n = \left(\frac{x}{n+1}\right)^n$$

$$(a_n)^{\frac{1}{n}} = \left(\left(\frac{x}{n+1}\right)^n\right)^{\frac{1}{n}} = \frac{x}{n+1}$$

$$\lim_{n \rightarrow \infty} (a_n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{x}{n+1} = 0$$

$\downarrow$
convergent.

$$\# \quad \left[\frac{n}{n+1}\right]^{n^2}$$

$$a_n = \left(\frac{n}{n+1}\right)^{n^2}$$

$$(a_n)^{\frac{1}{n}} = \left[\left(\frac{n}{n+1}\right)^{n^2}\right]^{\frac{1}{n}}$$

$$\Rightarrow \left(\frac{n}{n+1}\right)^n$$

$$\Rightarrow \frac{n^n}{\left(1 + \frac{1}{n}\right)^n}$$

$$\left(\frac{1}{1 + \frac{1}{n}}\right)^n$$

$$\frac{1}{e} = \frac{1}{2.7} < 1 \rightarrow \text{convergent}$$

Rabbe's test: if $\sum a_n$ be a series of positive term the series is convergent if

i) $\lim_{n \rightarrow \infty} n \left[\frac{a_n}{a_{n+1}} - 1 \right] > 1 \rightarrow \text{convergent.}$

ii) $\lim_{n \rightarrow \infty} n \left[\frac{a_n}{a_{n+1}} - 1 \right] < 1 \rightarrow \text{divergent.}$

iii) $\lim_{n \rightarrow \infty} n \left[\frac{a_n}{a_{n+1}} - 1 \right] = 1 \rightarrow \text{Rabbe's test fails.}$

$1 + \frac{x}{2} + \frac{x^2}{5} + \frac{x^3}{10} + \frac{x^4}{17} + \dots$

$$a_n = \frac{x^n}{n^2 + 1}$$

$$a_{n+1} = \frac{x^{n+1}}{(n+1)^2 + 1}$$

$$\frac{a_{n+1}}{a_n} = \frac{x^{n+1}}{(n+1)^2 + 1} \times \frac{n^2 + 1}{x^n}$$

$$\Rightarrow \frac{x \cdot n^2 + 1}{(n+1)^2 + 1}$$

$$\lim_{n \rightarrow \infty} \frac{x \cdot n^2 + 1}{n^2 + 2n + 2}$$

$$\Rightarrow x \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n^2}}{1 + \frac{2}{n} + \frac{2}{n^2}}$$

$$L = x \cdot \frac{1+0}{1+0+0} = x \cdot 1 = x.$$

$$x < 1 \rightarrow \text{convergent.}$$

$$x > 1 \rightarrow \text{divergent}$$

$$x = 1 \rightarrow \text{Ratio test fails}$$

$$\alpha = 1. \quad \frac{1^n}{n^2+1} \Rightarrow 1 + \frac{1}{2} + \frac{1}{5} + \frac{1}{10} \dots$$

$$\frac{u_n}{u_{n+1}} = \frac{n^2+2n+2}{n^2+1}$$

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left(\frac{n^2+2n+2}{n^2+1} - 1 \right)$$

$$\lim_{n \rightarrow \infty} n \left[\frac{n^2+2n+2 - n^2 - 1}{n^2+1} \right]$$

$$\lim_{n \rightarrow \infty} n \left(\frac{2n+1}{n^2+1} \right) \Rightarrow \lim_{n \rightarrow \infty} \frac{2n^2+n}{n^2+1}$$

$$\lim_{n \rightarrow \infty} \frac{2 + \frac{1}{n}}{1 + \frac{1}{n^2}} \Rightarrow \frac{2}{1} = 2.$$

$2 > 1 \rightarrow$ convergent.

$$\# \quad x^n \cdot \frac{n}{n^2+1}$$

$$a_n = \frac{x^n \cdot n}{n^2+1}$$

$$a_{n+1} = \frac{x^{n+1} (n+1)}{(n+1)^2+1}$$

$$\frac{a_{n+1}}{a_n} = \frac{x^{n+1} (n+1)}{(n+1)^2+1} \times \frac{n^2+1}{x^n}$$

$$\Rightarrow \left[\cancel{x} \cdot \cancel{a_{n+1}} \times \frac{n^2+1}{(n+1)^2+1} \right] \times$$

$$\Rightarrow x \cdot \frac{n+1}{n} \times \frac{n^2+1}{(n+1)^2+1}$$

$$x \lim_{n \rightarrow \infty} \left[\frac{n+1}{n} \times \frac{n^2+1}{(n+1)^2+1} \right]$$

$$\Rightarrow x \lim_{n \rightarrow \infty} \left[1 + \frac{1}{n} \times \frac{1 + \frac{1}{n^2}}{1 + \frac{2}{n} + \frac{2}{n^2}} \right]$$

$$x \times 1 = x.$$

$x < 1 \rightarrow$ convergent.

$x > 1 \rightarrow$ divergent.

$x = 1 \rightarrow$ Ratio test fails.

$$x = 1.$$

$$\frac{x^n \cdot n}{n^2+1} = \frac{n}{n^2+1} \Rightarrow u_n = \frac{n}{n^2+1}$$

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) > 1.$$

$$\frac{u_n}{u_{n+1}} = \frac{n}{n^2+1} \times \frac{(n+1)^2+1}{n+1}$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left[\frac{n(n^2+2n+2)}{(n+1)(n^2+1)} - 1 \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left[\frac{n^3+2n^2+2n - n^3 - n^2 - n - 1}{n^3+n^2+n+1} \right]$$

$$\lim_{n \rightarrow \infty} n \left[\frac{n^2+n-1}{n^3+n^2+n+1} \right]$$

$$\lim_{n \rightarrow \infty} \frac{n^3+n^2-0}{n^3+n^2+n+1}$$

$$\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n} - \frac{1}{n^2}}{1 + \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3}} = \frac{1+0-0}{1+0+0+0} = 1.$$

Rabbe's test fails.

$$\lim_{n \rightarrow \infty} \frac{a_n}{\sqrt[n]{n}} \Rightarrow \lim_{n \rightarrow \infty} \frac{n/n^2+1}{1/n}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n^2}} = 1.$$

Both series will behave same.

$\sum n$ is divergent.

$$\# (n^3+1)^{1/3} - n.$$

$$a_n = (n^3+1)^{1/3} - n.$$

$$\Rightarrow (n^3 [1 + \frac{1}{n^3}])^{1/3} - n.$$

$$\Rightarrow (n^3)^{1/3} (1 + \frac{1}{n^3})^{1/3} - n.$$

$$\Rightarrow n [1 + \frac{1}{n^3}]^{1/3} - n.$$

$$a_n = n \left[\left(1 + \frac{1}{n^3}\right)^{1/3} - 1 \right]$$

Using Binomial Expression $\rightarrow (1+x)^c$

$$x = \frac{1}{n^3} \quad c = 1/3.$$

$$(1+x)^c = 1 + cx + \frac{c(c-1)}{2!} x^2 + \dots$$

$$\left(1 + \frac{1}{n^3}\right)^{1/3} = 1 + \frac{1}{3} \left(\frac{1}{n^3}\right) + \frac{\frac{1}{3}(\frac{1}{3}-1)}{2!} \left(\frac{1}{n^3}\right)^2 + \dots$$

$$\left(1 + \frac{1}{n^3}\right)^{1/3} = 1 + \frac{1}{3n^3} + \frac{\frac{1}{3}(-\frac{2}{3})}{2} \frac{1}{n^6} - \dots$$

$$\left(1 + \frac{1}{n^3}\right)^{1/3} = 1 + \frac{1}{3n^3} - \frac{1}{9n^6} + \dots$$

$$a_n = n \left[\left(1 + \frac{1}{3n^3} - \frac{1}{9n^6} + \dots \right) - 1 \right]$$

$$\Rightarrow n \left[\frac{1}{3n^3} - \frac{1}{9n^6} + \dots \right]$$

$$\Rightarrow \frac{n}{3n^3} - \frac{n}{9n^6} + \dots$$

$$\Rightarrow \frac{1}{3n^2} - \frac{1}{9n^5} + \dots$$

$$v_n = \frac{1}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} \Rightarrow \lim_{n \rightarrow \infty} \frac{\frac{1}{3n^2} - \frac{1}{9n^5} + \dots}{\frac{1}{n^2}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left(\frac{1}{3n^2} - \frac{1}{9n^5} + \dots \right)$$

$$\Rightarrow \left(\frac{1}{3} - \frac{1}{9n^3} + \dots \right)$$

$$L = \frac{1}{3}$$

$\frac{1}{3} \rightarrow$ finite

Both series will behave same.

$v_n \rightarrow 2 > 1 \rightarrow$ convergent.

$$\# \quad \frac{1}{4} + \frac{1 \cdot 3}{4 \cdot 6} + \frac{1 \cdot 3 \cdot 5}{4 \cdot 6 \cdot 8} + \dots$$

$$a_n = \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{4 \cdot 6 \cdot 8 \dots (2n+2)}$$

$$a_{n+1} = \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{4 \cdot 6 \cdot 8 \dots (2n+4)}$$

$$\frac{u_{n+1}}{u_n} = \frac{2n+1}{2n+4}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{2n+1}{2n+4}$$

$$\lim_{n \rightarrow \infty} \frac{2 + \frac{1}{n}}{2 + \frac{4}{n}} = \frac{2}{2} = 1.$$

Ratio test fails.

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) > 1.$$

$$\frac{u_n}{u_{n+1}} = \frac{2n+4}{2n+1}$$

$$\lim_{n \rightarrow \infty} n \left[\frac{2n+4}{2n+1} - 1 \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left[\frac{(2n+4) - (2n+1)}{2n+1} \right]$$

$$\lim_{n \rightarrow \infty} n \left[\frac{3}{2n+1} \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{3n}{2n+1}$$

$$\lim_{n \rightarrow \infty} \frac{3}{2 + \frac{1}{n}} = \frac{3}{2} > 1 \rightarrow \text{Convergent.}$$

$$1 + \frac{1}{2}x + \frac{1 \cdot 3}{2 \cdot 4}x^2 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}x^3 - \dots$$

$$a_n = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} x^n.$$

$$a_{n+1} = \frac{1 \cdot 3 \cdot 5 \cdots (2n+1)}{2 \cdot 4 \cdot 6 \cdots (2n+2)} x^{n+1}.$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{1 \cdot 3 \cdot 5 \cdots (2n+1)}{2 \cdot 4 \cdot 6 \cdots (2n+2)} x^{n+1} \times \frac{2 \cdot 4 \cdot 6 \cdots (2n) x^n}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$$

$$\Rightarrow \frac{2n+1}{2n+2} x = \lim_{n \rightarrow \infty} x \frac{2n+1}{2n+2}$$

$$\Rightarrow x \lim_{n \rightarrow \infty} \frac{2 + \frac{1}{n}}{2 + \frac{2}{n}}$$

$$\Rightarrow x \cdot \frac{2}{2} = x.$$

$$\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1}$$

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left(\frac{2n+2}{2n+1} - 1 \right)$$

$$\lim_{n \rightarrow \infty} n \left[\frac{2n+2 - 2n-1}{2n+1} \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left(\frac{1}{2n+1} \right) \Rightarrow \lim_{n \rightarrow \infty} \frac{n}{2n+1}$$

$$\lim_{n \rightarrow \infty} \frac{1}{2 + \frac{1}{n}} = \frac{1}{2} < 1 \rightarrow \text{Divergent.}$$

Logarithm Test :- $\lim_{n \rightarrow \infty} n \log \frac{u_n}{u_{n+1}}$ be a positive term series if

$$\lim_{n \rightarrow \infty} n \log \frac{u_n}{u_{n+1}} > 1 \rightarrow \text{convergent.}$$

$$\lim_{n \rightarrow \infty} n \log \frac{u_n}{u_{n+1}} < 1 \rightarrow \text{divergent.}$$

$$\lim_{n \rightarrow \infty} n \log \frac{u_n}{u_{n+1}} = 1 \rightarrow \text{logarithm test fails.}$$

$$\# \quad x + \frac{2^2 x^2}{2!} + \frac{3^3 x^3}{3!} + \frac{4^4 x^4}{4!} \dots$$

$$a_n = \frac{n^n x^n}{n!}$$

$$a_{n+1} = \frac{(n+1)^{n+1} x^{n+1}}{(n+1)!}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{(n+1)^{n+1} x^{n+1}}{(n+1)!} \times \frac{n!}{n^n x^n}$$

$$\Rightarrow x \cdot \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{n^n} \times \frac{n!}{(n+1)!}$$

$$\Rightarrow x \cdot \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{n^n (n+1)}$$

$$\Rightarrow x \cdot \lim_{n \rightarrow \infty} \frac{(n+1)^n}{n^n}$$

$$\Rightarrow x \cdot \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n \Rightarrow x \cdot \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

$$\Rightarrow x \cdot e$$

$$x \cdot e < 1$$

$x < \frac{1}{e} \rightarrow$ convergent.

$x > \frac{1}{e} \rightarrow$ divergent.

$$\lim_{n \rightarrow \infty} n \log \left(\frac{u_n}{u_{n+1}} \right) > 1 \rightarrow \text{convergent}$$

$$\frac{u_n}{u_{n+1}} = \frac{1}{x \left(1 + \frac{1}{n} \right)^n} = \frac{1}{e \left(1 + \frac{1}{n} \right)^n} = e \left(1 + \frac{1}{n} \right)^{-n}$$

$$\lim_{n \rightarrow \infty} n \log \left[e \left(1 + \frac{1}{n} \right)^{-n} \right]$$

$$\lim_{n \rightarrow \infty} n \left[\log e + \log \left[\left(1 + \frac{1}{n} \right)^{-n} \right] \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left[1 - n \log \left(1 + \frac{1}{n} \right) \right]$$

$$\lim_{n \rightarrow \infty} n \left[1 - n \left(\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \dots \right) \right]$$

$$\lim_{n \rightarrow \infty} n \left[1 - \left(1 - \frac{1}{2n} + \frac{1}{3n^2} - \dots \right) \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} n \left[\frac{1}{2n} - \frac{1}{3n^2} \right]$$

$$\lim_{n \rightarrow \infty} \left[\frac{n}{2n} - \frac{n}{3n^2} - \dots \right]$$

$$\lim_{n \rightarrow \infty} \left[\frac{1}{2} - \frac{1}{3n} + \frac{1}{n^2} \right]$$

$$\frac{1}{2} - 0 + 0 = \frac{1}{2} < 1 \rightarrow \text{divergent.}$$

Taylor's Series: If a function $f(x)$ has a derivative of all orders at a point 'a' its Taylor's series centered at 'a' is $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$

Maclaurin Series: When $a=0$ in Taylor Series then it is called Maclaurin Series.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$\# f(x) = e^x$$

$$f(0) = 1$$

$$f'(x) = e^x$$

$$f''(x) = e^x$$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\# f(x) = \sin x$$

$$f(0) = 0$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

$$\Rightarrow 0 + x + 0 + \frac{x^3}{3!} + \dots$$

$$\Rightarrow x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$v = \frac{1}{\cos} = \frac{1}{\cos(0)} \text{ m/s} = \infty \text{ m/s}$$

$$\# f(x) = \cos x$$

$$f(0) = 1$$

$$f'(x) = -\sin x$$

$$f''(x) = -\cos x$$

$$f(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Alternative Series:- A Series in which +ve & -ve terms involve alternatively is called as alternating Series.

Leibnitz test:- If U_n is monotonically decreasing and limit is zero then the alternative series is convergent otherwise we can say that the series is divergent.

$\sum \frac{1}{1 \cdot 2} - \frac{1}{3 \cdot 4} + \frac{1}{5 \cdot 6} - \frac{1}{7 \cdot 8} + \dots$

$$a_n = \frac{(-1)^{n+1}}{(2n-1)(2n)}$$

$$a_{n+1} = \frac{1}{(2n+1)(2n+2)}$$

$$a_{n+1} < a_n \rightarrow \text{decreasing.}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{(2n-1)(2n)} = \frac{1}{\infty} = 0$$

Convergent

Power Series in 'x' term:- The Series $a_0 + a_1x + a_2x^2 + \dots + a_nx^n + \dots$ is power series in 'x' term $a_0, a_1, a_2, \dots, a_n$ are independent of 'x'.